

Water as a medium for microwave-assisted grafting of maleic anhydride onto a polypropylene film to improve its adhesion with a polyamide 6 film

Wen-Xin Jiang¹, Jin-Tang Duan², Lian-Fang Feng^{1,2}, Xue-Ping Gu^{1,2},

Cai-Liang Zhang^{1,2*}, Guo-Hua Hu^{3*}

¹State Key Laboratory of Chemical Engineering, College of Chemical and Biological Engineering, Zhejiang University, Hangzhou 310027, China

²Institute of Zhejiang University–Quzhou, 78 Jiu Hua North Boulevard, Quzhou 324000, China

³CNRS-Université de Lorraine, Laboratoire Réactions et Génie des Procédés (LRGP, UMR CNRS 7274), ENSIC, 1 rue Grandville, BP 20451, 54001 Nancy, France

*Correspondence concerning this article should be addressed to both C-L Zhang at zhangcailiang@zju.edu.cn and G-H Hu at guo-hua.hu@univ-lorraine.fr

Abstract: Maleic anhydride (MAH)-modified polypropylene (PP), denoted as PP-g-MAH, is often prepared by melt grafting of MAH onto PP through reactive extrusion. A major problem of this process is the presence of a large amount of residual (unreacted) MAH in the polymer, which is harmful for applications like food packaging. This work aims to study microwave-assisted grafting of MAH onto PP film surfaces using water instead of organic solvents as a medium to reduce MAH residues in PP. Microwave irradiation shortens the grafting time by accelerating the decomposition of initiator. Addition of a very small amount (2%) of xylene to water swells the PP film surfaces of about 12 μm thick to which the MAH grafting is confined. The adhesive strength of a double layer film based on PP-g-MAH (moisture barrier layer) and polyamide 6 (PA6, oxygen barrier layer) reaches 13.3 N/15mm instead of 2.4 N/15mm for that based on PP and PA6. A much less amount of the oil-soluble benzoyl peroxide is more efficient as an initiator for MAH grafting than water-soluble ammonium persulfate.

Keywords: Microwave, grafting, polypropylene, maleic anhydride, adhesive strength

1. Introduction

Food packaging is to protect food against humidity and oxygen threats. Polypropylene (PP) is commonly used as water-barrier layer in food packaging [1, 2] owing to its good hydrophobicity, mechanical properties, chemical and thermal stability. However, it is highly permeable to oxygen [3], and therefore should be combined with a good oxygen-barrier layer such as polyamide 6 (PA6) to form a multilayer film for food packaging [4]. Because PP, an apolar polymer, and PA6, a polar polymer, are highly incompatible, a tie-layer is often required to bound them together [5]. A common one is maleic anhydride (MAH) modified PP, denoted as PP-g-MAH. Free radical grafting of vinyl monomers such as MAH and glycidyl methacrylate onto polymer backbones such as polyolefins is initially conducted in solution. The use of solvents is concomitant with additional costs related to the solvents themselves and their recovery, and the negative impacts of the use of solvents on the environment. In order to avoid using organic solvents, it is now common to modify polymers in the melt by reactive extrusion, a green process with unique features [6, 7]. However, a major problem of this process is the presence of a large amount of residual (unreacted) monomers in the polymer [8, 9]. This may adversely affect the adhesive strength of multilayer films and food safety [10, 11].

Since the adhesion results from interaction between two polymer surfaces in contact, a tie-layer may not be necessary if the surface of a PP layer is chemically modified [12], reducing the issue of residual monomers [13, 14]. To do this, a key is to create reactive sites such as free radicals on the surface of the PP layer so that polar groups bearing vinyl monomers can be grafted onto through these reactive sites.

One way of creating reactive sites on a polymer surface for subsequent chemical modification is to subject it to radiation [13]. Such methods for PP surface modification include plasma [15], electron beam [16] and γ ray [17]. They have inherent drawbacks. Free radicals produced by plasma tend to react with its carrier gas such as oxygen to form new chemical structures on the PP surface, reducing the ability of the PP surface for subsequent grafting reaction with specific polar groups [18-20]. Electron beam and γ ray have very high energy and may strongly damage the PP surface. Indeed, such treatment may adversely affect the mechanical properties of PP like elastic modulus and tensile strength [21].

Unlike the above radiation methods, microwave is more selective in the sense that it can only be effectively absorbed by polar molecules and not by apolar ones [22]. It can be used to advantage for the grafting of MAH (polar) onto PP (apolar) film surfaces initiated by a peroxide. Microwave irradiation promotes the molecular vibration of the peroxide and therefore accelerates its decomposition rate [23, 24]. Thus, the grafting reaction can proceed more rapidly without adversely affecting the mechanical properties of PP [25-27]. Recently, microwave is used to assist the free radical grafting of MAH onto the PP surface in order to improve the adhesion performance of the latter with other substrates [14, 28, 29]. In this process, a PP film is immersed in a polar organic solvent like *N, N*-dimethylformamide (DMF) capable of absorbing microwave irradiation. However, a polar solvent cannot swell PP surfaces much, limiting the extent of MAH grafting. When a non-polar organic solvent like xylene is used, it better swells the PP surfaces, promoting MAH grafting and consequently the adhesion performance of the PP with PA6 [14]. However, the use of a large amounts of organic solvents, whether they are polar

or apolar, faces issues like environmental pollution, toxicity, and solvent separation and recovery.

In this work, a mixture of water and xylene (only a few percent with regard to water) is used as a reaction medium to substitute for full organic solvents for microwave-assisted free radical grafting of MAH onto PP film surfaces. Water can be heated up very rapidly by microwave and xylene which is little soluble in water tends to be located on and swell the PP film surfaces. Both effects are expected to favor the MAH grafting on the PP film surfaces. The mechanism of the MAH grafting reaction in the mixture of water and xylene is discussed. The effects of various reaction parameters on the adhesion between PP and PA6 layers are quantified.

2. Experimental methods

2.1 Materials

A PP film and a PA6 film of 0.2 mm thick were obtained from Zhejiang Wazam New Materials Co., Ltd., China. Maleic anhydride (MAH, AR), xylene (AR), DMF (AR), styrene (St, AR), acetone (AR), chloroform (CHCl_3 , AR) and methanol (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd., China. Oil red O ($\text{C}_{26}\text{H}_{24}\text{N}_4\text{O}$, AR), ammonium persulfate (APS, purity $\geq 98\%$), benzoyl peroxide (BPO, containing 25% water) and alkaline aluminum oxide powder (200-300 mesh) were purchased from J&K Scientific Ltd., China. Prior to experiments, St passed through a column filled with alkaline aluminum oxide powder to remove inhibitors. BPO was purified by recrystallization in a mixture of CHCl_3 and methanol with a ratio of 1:3.

2.2 Microwave-assisted free radical grafting of MAH onto a PP film in water

Fig. 1 shows the experimental procedure for the microwave-assisted free radical grafting of MAH onto the PP film. A beaker containing a mixture of water (200mL) and appropriate amounts of MAH and xylene (a swelling agent for the PP) was heated to a given temperature (in the range of 65 to 80°C). A piece of the PP film (12cm × 5cm × 0.2mm) was immersed in the above mixture and then APS (a water soluble free radical initiator) was added in. The mixture was stirred by a glass rod for about 10s so that xylene wet and swell the PP film surfaces. The beaker was put in a microwave chamber (model FCMCR-3SX-T, Kerui Instrument Inc., China) so that the grafting reaction proceeded at the given reaction temperature for a chosen period of time. After the reaction, the MAH-modified PP film was immersed in acetone for 12h and then rinsed with acetone 3 times to remove MAH and xylene residues. It was dried in a vacuum oven at 90°C for 1h.

2.3 Measurements

2.3.1 X-ray photoelectron spectra analysis (MAH modified or not)

X-ray photoelectron spectrometer (XPS, model Escalab250Xi, ThermoFisher Scientific Inc., USA) was used for elemental composition analysis of the PP film surface. The energy analysis range was 0-5000 eV and the energy step was 6 meV. Al K α was used as the X-ray source. The bonding energy was calibrated by C 1s peak of 284.4 eV.

2.3.2 Surface morphology of the PP film (MAH modified or not)

The surface morphology of the PP film was characterized by scanning electron microscopy

(SEM, model SU-3500, Hitachi, Japan) with an acceleration voltage of 30 kV. Prior to scanning, one of the PP film surfaces was sprayed with a layer of gold-palladium alloy.

2.3.3 Water contact angle on the PP film (MAH modified or not)

A video optical contact angle measuring instrument (model OCA 20, Dataphysics, German) was used to measure the contact angle of water on the PP film at room temperature. The average of at least 5 measurements was taken as the contact angle.

2.3.4 Adhesive strength between PP and PA6 double layer films

Two films (PP and PA6) of the same size were compressed together using a hot press (model GT-7014-A50C, Gotech Testing Machine Inc., China) at 220°C and 4MPa for 40s. The resulting double layer PP/PA6 film was cut to pieces of 100mm × 15mm × 0.2mm for T-peel testing (model Zwick/Roell Z020, Zwick, German) with a peeling rate of 10mm/min. The maximum stress measured during the test was taken as the adhesive strength (unit: N/15mm). The average of at least 3 tests was taken as the adhesive strength.

2.3.5 Measurement of the tensile properties of the PP film

The tensile properties of the PP film before and after the grafting reaction were measured and compared. It was punched into a dumbbell shape (overall length: 35 mm, width of the parallel part: 2 mm, thickness: 0.2 mm). The tensile machine used was Zwick/Roell Z020, Zwick, Germany. At least three measurements were done for each sample.

3. Results and discussion

This work aims to improve the adhesive strength of a PP film with a PA6 film by microwave-assisted free radical grafting of MAH onto the PP film surfaces. Water is used as the

grafting reaction medium to substitute for an organic substance such as DMF or xylene which are toxic. In order for MAH to be grafted on the PP surfaces, all the necessary reactive agents (MAH and free radical initiator) have to penetrate them to a certain extent (of the order of a few to a dozen of micrometers for example). Thus, it is conceivable that water alone as a medium is likely unfavorable for the MAH grafting, as MAH and/or the free radical initiator in water cannot penetrate the PP surfaces to a sufficient extent. In order to overcome this issue, a small amount of xylene is added to water. It is capable of swelling without (significantly) dissolving the PP surfaces in a certain range of temperature (below 85°C), offering MAH and the free radical initiator a possibility to penetrating the swollen PP surfaces and then react. It is therefore important to check whether or not the presence of xylene in water is able to swell the PP surfaces, to assess the swelling process and to establish a relationship between the swollen/penetration depth of the PP film (12cm × 5cm × 0.2mm) and the amount of xylene in water.

3.1 Visualization of the swelling process of the PP film surfaces by xylene

Xylene has a very small solubility in water (145 to 185 mg/L water at 25°C) and is lighter than water. When it is added to water in a beaker, it becomes the supernatant under static conditions, as shown in Fig. 1. When a PP film is immersed in a phase-separated mixture of water and xylene (in a small amount), it is conceivable that its swelling by xylene would hardly proceed. This means that stirring is necessary to disperse xylene in water in order to promote the contact between xylene droplets and the PP film and consequently the swelling of the latter.

To visualize the swelling process of the PP film in water by xylene, oil red O is added to xylene as a dye. It is soluble in xylene and insoluble in water. Fig. 2 shows the dyeing and swelling process. When a very small amount of dyed xylene (5mL) is added to water (200mL) at 80°C, it stays on the water surface, as shown in (a) and (b). Upon the immersion of the PP film under stirring with a glass rod, xylene is quickly (on the order of 10s) absorbed onto the PP film surfaces (c). When the PP film is taken out of the mixture of water and xylene, most xylene is taken out together with the PP film (d) because of the affinity between them. Moreover, wrinkles appear on the PP film because of the swelling by xylene (e).

The very low solubility of xylene in water and lower density than water endow it with two important advantages. The absorption of xylene by the PP film is more rapid and more complete than if its solubility in water was high. It is easy to separate and recover xylene from water owing to its low solubility in water and lower density than water. To confirm these two advantages, the swelling of the PP film by xylene in DMF is compared with that in water. When xylene dyed with oil red O (5mL) is added to DMF (200mL), the entire mixture becomes red as they both are miscible (Fig. 3a). When immersed in the above mixture, the PP film is no longer dyed nor swollen (Fig. 3b), unlike in the mixture of water and xylene. This is because xylene now has to be partitioned between DMF and PP film. The small amount of xylene (5mL) in DMF (200mL) is no longer enough to swell the PP film.

3.2 Extent of the swelling of the PP film surfaces by xylene

Fig. 4 compares the PP film surfaces and the contact angles of water on them after microwave-assisted MAH grafting in water without and with xylene as a swelling agent of the PP film. APS is used as a free radical initiator and is water-soluble. The water contact angle on a PP film modified by microwave-assisted MAH grafting in water without xylene is 103.6° , which is almost the same as that of the initial PP film (104.1°). When the microwave-assisted MAH grafting is conducted in water in the presence of xylene (xylene/water ratio: 1/50), the water contact angle is significantly reduced from 104.1° (the initial PP film) to 79.8° (PP-g-MAH obtained in the presence of xylene). Moreover, there appears pores on the PP-g-MAH surfaces. These results confirm that xylene is indeed a swelling agent for the PP film.

Fig. 5 shows the xylene penetration (swollen) depth in the thickness direction of the PP film after 30 min of immersion in a mixture of water and xylene at 80°C . It is about 6, 12 and $12\mu\text{m}$ when the xylene/water ratio is 1/100, 1/50 and 1/33, respectively. This indicates that at 80°C , the xylene penetration depth first increases with increasing xylene/water volume ratio, and starts to level off at about $12\mu\text{m}$ (the PP film is $200\mu\text{m}$ thick) when the xylene/water ratio is 1/50 or higher.

After the grafting reaction, xylene and MAH residue or its derivatives may remain within a certain depth (about $12\mu\text{m}$) of the PP-g-MAH surfaces. They may be harmful for food packaging and need to be removed. This is done by washing the PP-g-MAH film in acetone. Gas chromatography analysis shows that the immersion of the PP-g-MAH in acetone for 5 min or longer allows to remove all xylene and/or MAH from it (see supporting information). Moreover, since the microwave-induced grafting reaction occurs only within the swollen depth (about 12

μm) of the PP film surfaces, its mechanical properties are expected to be damaged to a much less extent compared to the MAH grafting in the PP film bulk. Indeed, the tensile strength and elongation at break of the PP film of 200 μm thick (59.1 ± 5.3 MPa and $500\pm8\%$, respectively) are very close to those of the PP-g-MAH obtained by the microwave-induced grafting (56.7 ± 4.7 MPa and $518\pm14\%$, respectively). Nevertheless, if the PP film is much thinner, say, 50 μm , the above conclusion may not necessarily hold.

3.3 Effects of reaction conditions on the adhesive strength

As shown in Fig. 6, compared with the virgin PP, the XPS spectrum of the modified PP shows a new but very small peak at 288.08 eV [30], which corresponds to the O-C=O bond of MAH. This indicates that MAH is grafted onto the PP film surface. However, this peak is too small for further quantitative exploitation. The very small peak also suggests that the grafting of MAH only proceeds on the PP film surface. This explains the fact that the FTIR and ATR-FTIR analyses cannot distinguish between the virgin PP and the MAH modified one. Thus, it is assessed indirectly through the adhesive strength between the PP and PA6 layers, with the assumption that a high adhesive strength corresponds to a high amount of the grafted MAH on the PP layer surface. In what follows, the effects of various reaction parameters on the adhesive strength of the PP-g-MAH are discussed.

3.3.1 Effect of microwave power

MAH is grafted onto the PP film ($12\text{cm} \times 5\text{cm} \times 0.2\text{mm}$) in the mixture of water (200mL), xylene (5mL), MAH (6g) and APS (0.6g) with various levels of microwave power. From Fig. 7,

the adhesive strength of the PP-g-MAH without microwave is 5.6N/15mm. It increases with increasing microwave power and reaches a plateau of about 12N/15mm when the microwave power is 720W or higher. This suggests that microwave does significantly promote the grafting of MAH onto the PP film surfaces.

3.3.2 Effect of the presence and amount of xylene in water

MAH is grafted onto the PP film in water without or with xylene. From Fig. 8, the adhesive strength between the PP-g-MAH obtained without xylene (xylene/water volume ratio = 0) and the PA6 is 2.4N/15mm. It increases very significantly when a very small amount of xylene is added to water, and reaches a plateau of 13.1N/15mm when the xylene/water volume ratio is 1/50. This indicates that a very small amount of xylene is both necessary and sufficient to wet and swell a PP film of 12cm × 5cm × 0.2mm (in agreement with the results in Fig. 5) so that the MAH grafting can proceed to a sufficient extent on the PP surfaces to endow it with a high adhesive strength. When the xylene/water ratio is above 1/50, the adhesive strength levels off or seems to slightly decrease. As discussed later, a high amount of xylene may erode the PP film surfaces to an extent that is detrimental to their adhesion with the PA6. Also, an excess of xylene that is not absorbed by the PP film remains in water and may capture some MAH. This is detrimental to the grafting of MAH onto the PP film surfaces.

The results in Fig. 8 are in agreement with those in Fig. 9. The latter shows that when the xylene/water ratio is 1/100 at 80°C, the amount of xylene is not enough to fully cover the PP film of 12cm × 5cm × 0.2mm. After the PP film is taken out of the beaker, the water is colorless. This indicates that all xylene is absorbed by the PP film. At a higher xylene/water ratio of 1/50, a

limited number of red color spots are visible on the water surface, indicating that this amount of xylene is enough to fully cover the PP film and is almost completely absorbed by the latter. When the xylene/water ratio reaches 1/33, the PP film can no longer absorb all xylene, as indicated by a significant number of red spots on the water surface after the PP film is taken out of the beaker. In what follows, the xylene/water volume ratio is set at 1/40 (slightly higher than 1/50) so that the amount of xylene is always enough to cover the PP film.

3.3.3 Effect of temperature

Fig. 10 shows that for a given reaction time (30min), the adhesive strength first increases with increasing temperature from 65 to 75°C and then starts to level off when the temperature is 80°C or higher. A higher temperature results in a higher extent of swelling of the PP film, on the one hand, and a more rapid formation of free radicals, on the other hand. Both effects are beneficial for the MAH grafting and consequently the adhesive strength. However, the temperature should not exceed 85°C, otherwise the integrity of the PP film could be damaged by excessive swelling/dissolution due to xylene.

The results in Fig. 10 are consistent with the amounts of the absorbed xylene by the PP film at the various temperatures. From Fig. 11, as the temperature increases, the water surface becomes less and less colored after the PP film is taken out from the breaker while the PP film is more and more colored. This indicates that the amount of the absorbed xylene by the PP film increases with increasing temperature till 80°C.

3.3.4 Effect of reaction time

Fig. 12 shows that for a given reaction temperature (80°C), the adhesive strength first

increases with increasing reaction time till around 10min and then starts to level off thereafter. This indicates that 30min is an appropriate time for the MAH grafting at 80°C.

3.3.5 Effect of using an oil-soluble free radical initiator

The results reported above are obtained using APS as a free radical initiator. It is water-soluble and is dissolved in water together with MAH for the MAH grafting. In these cases, MAH (6g) and APS (0.6g) are added to water (200mL). They must diffuse through the water phase to reach the xylene-swollen PP film surfaces so that the MAH grafting can proceed. One may also envision to use an oil-soluble free radical initiator such as BPO. In this case, a PP-g-MAH can be obtained in the following manner: (1) dissolution of MAH (6.0g) in water (200 mL) and BPO (0.01g) in xylene (5mL); (2) immersion of a PP film in the above solution of MAH and water, (3) addition of the above solution of BPO and xylene under stirring, (4) grafting under microwave. Fig. 13 compares the PP-g-MAH obtained by the BPO grafting system and that obtained by the APS grafting system (6.0g of MAH and 0.6g of APS dissolved in 200mL of water, plus 5mL of xylene) in different grafting times in terms of the adhesive strength with the PA6 film. After 10min of grafting reaction or longer under the microwave (900 W), both the oil-soluble BPO and water-soluble APS grafting systems lead to the same level of adhesive strength. However, the difference in the concentration of the initiator is huge: only 0.01 g for BPO and 0.6 g for APS. Moreover, the BPO grafting system reaches the plateau more rapidly, even though its half-life is even longer than that of APS. The fact that a much smaller amount of BPO than APS yields the same level of adhesive strength and the grafting time is shorter could be explained as follows. APS is only soluble in water (a large amount). Thus, itself and/or its free radicals must diffuse

through the water phase to reach PP film surfaces to initiate the grafting reaction. During the diffusion process, some free radicals may collide and react each other, losing their initiation ability for MAH grafting. By contrast, BPO is only soluble in xylene. As such, it is absorbed directly by the PP film where MAH grafting should take place.

Fig. 14 shows the effect of BPO dissolved in xylene (5mL) on the adhesive strength. A maximum adhesive strength of 11.8N/15mm is found for a BPO/xylene ratio of 0.002g/mL. Mechanisms leading to this maximum are unclear. However, several hypotheses can be made. One of them could be that as the BPO load increases, more free radicals will attack the PP film surfaces. When the molecular degradation of the PP film surfaces reaches a certain level, the adhesive strength will deteriorate.

3.3.6 Effect of adding styrene (St)

It is well-documented that when a vinyl monomer such as MAH and glycidyl methacrylate is grafted onto a polymer backbone such as PP by a free radical mechanism, the amount of grafted monomer is low (normally < 1 wt.% with respect to PP) and the molecular weight of the PP can be largely reduced. To overcome these two problems, the concept of comonomer is put forth [31-33]. It consists in adding a second vinyl monomer, called co-monomer. This concept works efficiently only when the following two conditions are met at the same time: (a) the comonomer reacts with a macroradical of the polymer backbone in a much faster manner than the target monomer; (b) once the comonomer has reacted with the macroradical of the polymer backbone, the resulting free radical can rapidly copolymerize with the target monomer. When a comonomer is used, the target monomer is no longer directly grafted onto the polymer backbone,

but through the reaction with the comonomer which is already grafted onto the polymer backbone.

St is added to the solution of xylene (5mL) and BPO (0.02g). The whole system is then added to water (200mL) containing MAH (6.0g). Fig. 15 shows the effect of the amount of St in xylene on the adhesive strength between the PP-g-MAH and PA6 films. The adhesive strength reaches a maximum of 13.3N/15mm when the St/xylene ratio is 0.04g/mL. This indicates that adding an appropriate amount of St as comonomer does further improve the adhesive strength. Nevertheless, when its amount exceeds a certain level, it starts to have an adverse effect on the adhesive strength. However, mechanisms leading to this maximum are unclear. One of the assumptions would be that when the amount of St exceeds a certain level, a large amount of copolymer of St and MAH would be formed. This disfavors the MAH grafting and consequently the adhesive strength.

4. Conclusion

This work uses water in the presence of a very small amount of xylene as a swelling agent to conduct microwave-induced grafting reaction of maleic anhydride (MAH) onto the surfaces of a polypropylene (PP) film in order to reduce negative impacts of using a large amount of organic solvents. Specifically, the presence of xylene in water (xylene/water: 1/50) at 80°C quickly swells a PP film with a depth of about 12.3 μm , providing the PP film with more volume for MAH grafting. Both water-soluble ammonium persulfate (APS) and benzoyl peroxide (BPO) are used as free radical initiator for the microwave-induced grafting of MAH onto a PP film. The

adhesive strength between a resulting PP-g-MAH film and a polyamide 6 (PA6) film can reach 13.3 N/15mm (it is only 2.4 N/15mm between the initial PP film and the PA6 film). A much smaller amount of BPO than APS can yield a PP-g-MAH the same level of adhesive strength owing to its much higher solubility in xylene. Addition of a very small amount of styrene can be used as comonomer to further increase the grafting of MAH onto the PP film.

CRedit Authorship Contribution Statement

Wen-Xin Jiang: Software, Formal analysis, Investigation, Writing-original draft. **Jin-Tang Duan:** Methodology, Data curation. **Lian-Fang Feng:** Writing-review and editing. **Xue-Ping Gu:** Writing-review and editing. **Cai-Liang Zhang:** Conceptualization, Supervision, Funding acquisition, Writing-review & editing. **Guo-Hua Hu:** Writing-review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could inappropriately influence the work reported in this paper.

Acknowledgement

The authors thank the National Natural Science Foundation of China (51973193), Key Research and Development Program of Zhejiang Province (2019C01100) and the Institute of Zhejiang University-Quzhou (IZQ2020KJ2005) for their financial support.

References

- [1] K.Q. Hu, Z.Y. Huyan, S.X. Ding, Y.Y. Dong, X.Z. Yu, Investigation on food packaging

- polymers: Effects on vegetable oil oxidation, *Food Chem.* 315 (2020) 126299, <https://doi.org/10.1016/j.foodchem.2020.126299>.
- [2] M. Kowalska, S. Janas, W. Kudelka, Consumer preferences and physical evaluation of shortbread stored in different packaging, *Int. J. Consum. Stud.* 44 (2020) 469-480, <https://doi.org/10.1111/ijcs.12579>.
- [3] P. Cerruti, M. Malinconico, J. Rychly, L. Matisova-Rychla, C. Carfagna, Effect of natural antioxidants on the stability of polypropylene films, *Polym. Degrad. Stab.* 94 (2009) 2095-2100, <https://doi.org/10.1016/j.polymdegradstab.2009.07.023>.
- [4] Y. Wang, W.J. Sun, S. Liu, H.J. Ji, X. Chen, H.B. Zhu, H.L. Zhao, Y.L. Ma, L.S. Xie, The formation of a highly oriented structure and improvement of properties in PP/PA6 polymer blends during extrusion-stretching, *Polymers* 12 (2020) 878, <https://doi.org/10.3390/polym12040878>.
- [5] K. Kaiser, M. Schmid, M. Schlummer, Recycling of polymer-based multilayer packaging: A review, *Recycling* 3 (2018) 1, <https://doi.org/10.3390/recycling3010001>.
- [6] T.T. Li, L.F. Feng, X.P. Gu, C.L. Zhang, P. Wang, G.H. Hu, Intensification of Polymerization Processes by Reactive Extrusion, *Ind. Eng. Chem. Res.* 60 (2021) 2791-2806, <https://doi.org/10.1021/acs.iecr.0c05078>.
- [7] G.H. Hu, H. Cartier, Free radical grafting of glycidyl methacrylate onto PP in a twin screw extruder: Influence of feeding mode, *Int. Polym. Process.* 13 (1998) 111-117, <https://doi.org/10.3139/217.980111>.
- [8] G.H. Hu, Y.J. Sun, M. Lambla, Devolatilization: A critical sequential operation for in-situ

- compatibilization of immiscible polymer blends by one-step reactive extrusion, *Polym. Eng. Sci.* 36 (1996) 676-684, <https://doi.org/10.1002/pen.10455>.
- [9] G.H. Hu, Y.J. Sun, M. Lambla, Effects of processing parameters on the in situ compatibilization of polypropylene and poly(butylene terephthalate) blends by one-step reactive extrusion, *J. Appl. Polym. Sci.* 61 (1996), 1039-1047, [https://doi.org/10.1002/\(SICI\)1097-4628\(19960808\)61:6<1039::AID-APP18>3.0.CO;2-R](https://doi.org/10.1002/(SICI)1097-4628(19960808)61:6<1039::AID-APP18>3.0.CO;2-R).
- [10] CEN/TS 13130-24-2005, Materials and articles in contact with foodstuffs-Plastics substances subject to limitation-Determination of maleic acid and maleic anhydride in food simulants, European Committee for Standardization, 2005.
- [11] P.J. Fordham, J.W. Gramshaw, L. Castle, Analysis for organic residues from aids to polymerization used to make plastics intended for food contact, *Food Addit. Contam.* 18 (2001) 461-471, <https://doi.org/10.1080/02652030010022520>.
- [12] J. Balart, V. Fombuena, J.M. Espana, L. Sanchez-Nacher, R. Balart, Improvement of adhesion properties of polypropylene substrates by methyl methacrylate UV photografting surface treatment, *Mater. Des.* 33 (2012) 1-10, <https://doi.org/10.1016/j.matdes.2011.06.069>.
- [13] M.M. Nasef, O. Güven, Radiation-grafted copolymers for separation and purification purposes: Status, challenges and future directions, *Prog. Polym. Sci.* 37 (2012) 1597-1656, <https://doi.org/10.1016/j.progpolymsci.2012.07.004>.
- [14] W.X. Jiang, L.F. Feng, X.P. Gu, J.T. Duan, C.L. Zhang, Microwave-induced grafting of maleic anhydride onto the surface of a polypropylene film for high adhesion to polyamide

- 6, *Ind. Eng. Chem. Res.* 60 (2021) 17556-17563, <https://doi.org/10.1021/acs.iecr.1c03701>.
- [15] J. Prachar, I. Novak, A. Kleinova, P. Gemeiner, M. Omastova, S. Chmela, E. Borsig, Plasma grafting of polypropylene with organosilanes and its alkylamine treatment, *Vacuum* 127 (2016) 38-44, <https://doi.org/10.1016/j.vacuum.2016.02.007>.
- [16] K. Hisada, M. Matsuoka, I. Tabata, K. Hirogaki, T. Hori, Two-step radical grafting onto polypropylene fiber initiated by active species prepared through the irradiation of electron beam, *J. Photopolym. Sci. Technol.* 26 (2013) 277-282, <https://doi.org/10.2494/photopolymer.26.277>.
- [17] M.I. ul Hassan, S. Taimur, I.A. Khan, T. Yasin, S.W. Ali, Surface modification of polypropylene waste by the radiation grafting of styrene and upcycling into a cation-exchange resin, *J. Appl. Polym. Sci.* 136 (2019) 47145, <https://doi.org/10.1002/app.47145>.
- [18] J. Friedrich, Plasma, in: J. Friedrich, *The plasma chemistry of polymer surfaces: advanced techniques for surface design*, Wiley-VCH, Weinheim, 2012, pp. 35-53.
- [19] L.H. Sharpe, Interfaces, Interphases and “Adhesion”: A Perspective, in: G. Akovali, *The interfacial interactions in polymeric composites*, Springer, Dordrecht, 1993, pp. 1-20.
- [20] F. Awaja, M. Gilbert, G. Kelly, B. Fox, J.P. Paul, Adhesion of polymers, *Prog. Polym. Sci.* 34 (2009) 948-968, <https://doi.org/10.1016/j.progpolymsci.2009.04.007>.
- [21] C. Burgstaller, T. Höftberger, B. Gallnböck-Wagner, W. Stadlbauer, Effects of radiation type and dose on the properties of selected polymers, *Polym. Eng. Sci.* 61 (2020) 39-54, <https://doi.org/10.1002/pen.25553>.
- [22] D.M.P. Mingos, D.R. Baghurst, Applications of microwave dielectric heating effects to

- synthetic problems in chemistry, *Chem. Soc. Rev.* 20 (1991) 1-47, <https://doi.org/10.1039/cs9912000001>.
- [23] B.T. Ergen, M. Bayramoglu, Kinetic approach for investigating the "microwave effect": decomposition of aqueous potassium persulfate, *Ind. Eng. Chem. Res.* 50 (2011) 6629-6637, <https://doi.org/10.1021/ie200095y>.
- [24] Z.D. Liu, Y.C. Lu, B.D. Yang, G.S. Luo, Free radical polymerization of butyl acrylate in monodispersed droplets: Comparison between two heating strategies, *J. Appl. Polym. Sci.* 127 (2012) 628-635, <https://doi.org/10.1002/app.37832>.
- [25] L. Zhang, H. Liu, X.Y. Qian, F.H. Liu, J. Zhang, Preparation of a novel oriented structure in carbon black filled isotactic polypropylene composites by microwave selective heating, *J. Macromol. Sci. B* 51 (2012) 1596-1605, <https://doi.org/10.1080/00222348.2012.657105>.
- [26] S.H. Wang, X.H. Zhang, Z.Y. Guo, H.B. Jiang, J.L. Qiao, Design and preparation of superwetting polymer surface, *Polymer* 186 (2020) 122043, <https://doi.org/10.1016/j.polymer.2019.122043>.
- [27] Y.N. Zhou, J.J. Li, Y.Y. Wu, Z.H. Luo, Role of external field in polymerization: mechanism and kinetics, *Chem. Rev.* 120 (2020) 2950-3048, <https://doi.org/10.1021/acs.chemrev.9b00744>.
- [28] S.H. Wang, X.H. Zhang, C. Jiang, H.B. Jiang, Y.J. Tang, J. Li, M.Q. Ren, J.L. Qiao, Polymer solid-phase grafting at temperature higher than the polymer melting point through selective heating, *Macromolecules* 52 (2019) 3222-3230,

<https://doi.org/10.1021/acs.macromol.8b02737>.

- [29] Y.Z. Wang, X.X. Zheng, Z.F. Chen, C.Y. Zheng, Microwave assisted radical grafting of maleic anhydride onto polyethylene in solution, *J. Macromol. Sci. A* 40 (2003) 739-745, <https://doi.org/10.1081/MA-120021423>.
- [30] C.S. Liang, Z.F. Lv, Y. Bo, J.Y. Cui, S.A. Xu, Effect of modified polypropylene on the interfacial bonding of polymer–aluminum laminated films. *Mater. Des.* 81 (2015) 141-148, <https://doi.org/10.1016/j.matdes.2015.05.021>.
- [31] H. Cartier, G.H. Hu, Plastification or melting: A critical process for free radical grafting in screw extruders, *Polym. Eng. Sci.* 38 (1998) 178-185, <https://doi.org/10.1002/pen.10178>.
- [32] Y.J. Sun, G.H. Hu, M. Lambla, Free radical grafting of glycidyl methacrylate onto polypropylene in a co-rotating twin screw extruder, *J. Appl. Polym. Sci.* 57 (1995) 1043-1054, <https://doi.org/10.1002/app.1995.070570903>.
- [33] Y.J. Sun, G.H. Hu, M. Lambla, Melt free radical grafting of glycidyl methacrylate onto polypropylene, *Angew. Makromol. Chem.* 229 (1995) 1-13, <https://doi.org/10.1002/apmc.1995.052290101>.

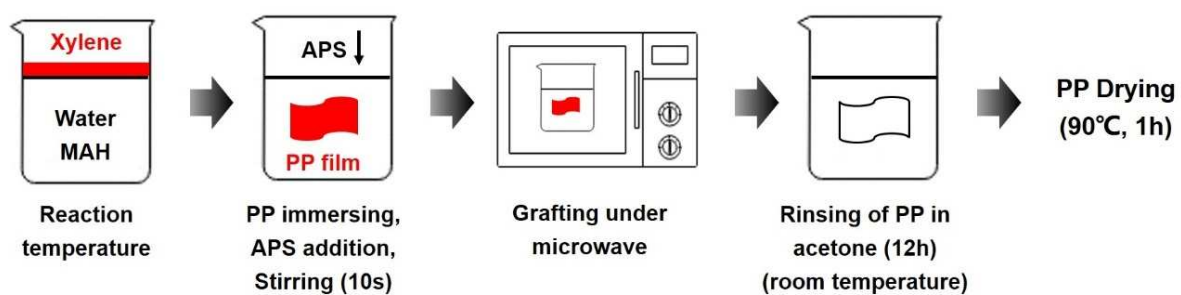


Fig. 1. Procedure of the microwave-assisted free radical grafting of MAH onto a PP film in water in the presence or absence of xylene.

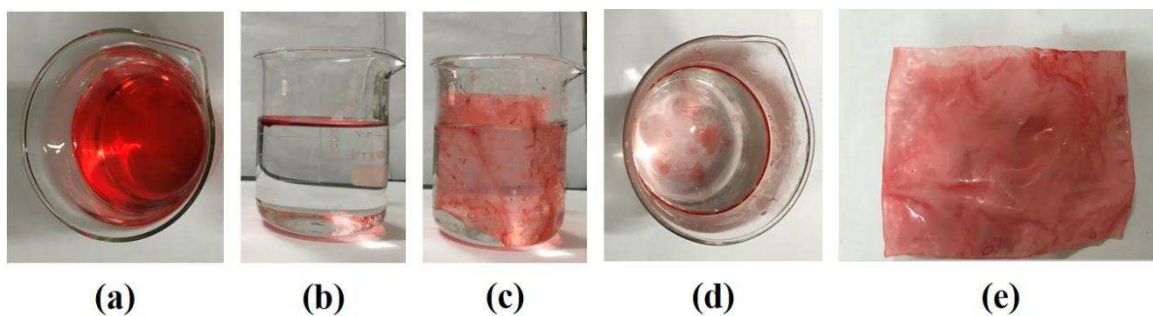


Fig. 2. Photographs of the dyeing and swelling process of a PP film ($12\text{cm} \times 5\text{cm} \times 0.2\text{mm}$) in a mixture of xylene and water (xylene/water ratio: 1/40) at 80°C . (a) and (b): top view and front view of the mixture of dyed xylene and water before the immersion of the PP film, respectively, (c): front view of the PP film immersed in the mixture of dyed xylene and water, (d): top view of the mixture of dyed xylene and water after the PP film is taken out, and (e) dyed PP film by dyed xylene.

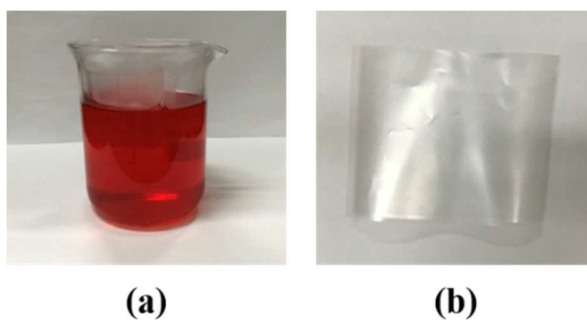


Fig. 3. Photographs of the dyeing and swelling process of the PP film in a mixture of xylene and DMF (1/40) at 80°C. (a) Front view of the solution of dyed xylene and DMF with the immersion of the PP film, (b) PP film taken out of the solution showing the absence of dyeing.

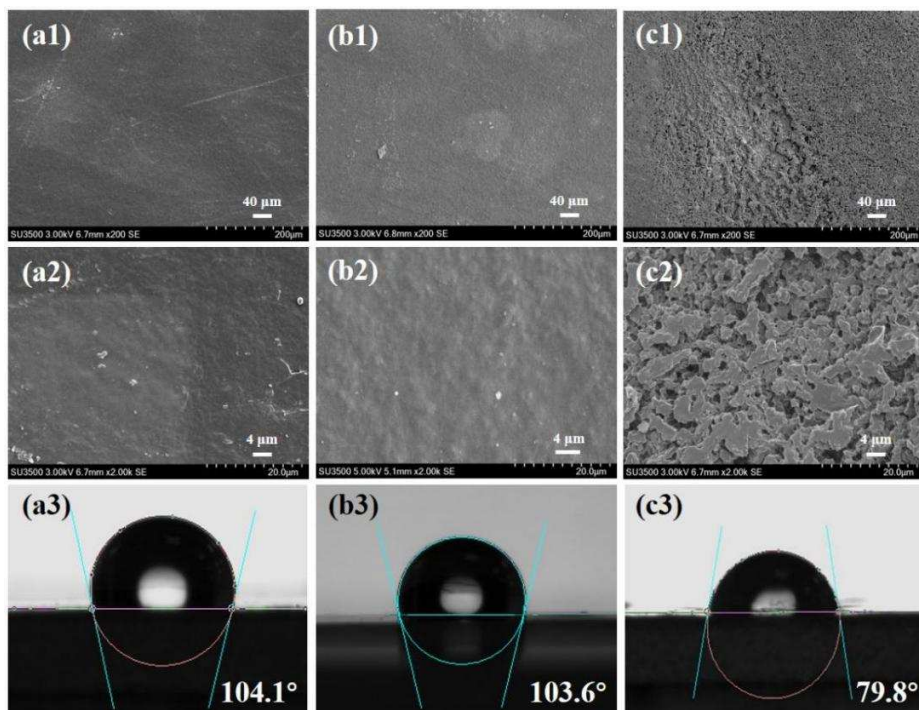


Fig. 4. SEM images showing the effects of the presence of xylene in water on the state of the PP film surfaces after the microwave-assisted MAH grafting. Microwave power: 900 W, temperature: 80°C, time: 30min, MAH: 6g, APS: 0.6g. (a1, a2) an original PP film surface, (b1, b2) a PP-g-MAH film surface obtained after the microwave-assisted grafting in water without xylene, (c1, c2) a PP-g-MAH film surface obtained after the microwave-assisted grafting in water with xylene (xylene/water ratio: 1/50). Magnification of the SEM images: 200 for (a1, b1, c1), and 2000 for (a2, b2, c2). (a3, b3, c3): contact angles of water on the PP film surfaces a1, b1 and c1, respectively.

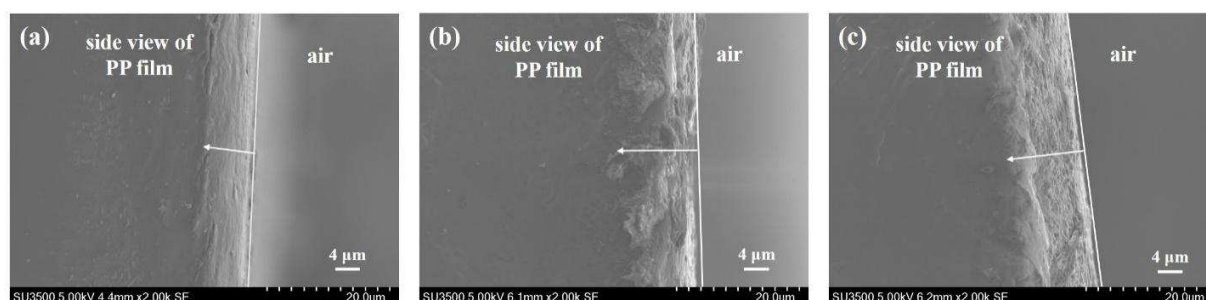


Fig. 5. Effect of the xylene/water volume ratio on the xylene penetration (swollen depth) in the thickness direction of the PP-g-MAH film obtained by microwave-assisted grafting in the mixture of xylene and water at 80°C for 30min. Xylene/water volume ratio: (a) 1/100, (b) 1/50 and (c) 1/33. The arrows indicate the direction of xylene penetration.

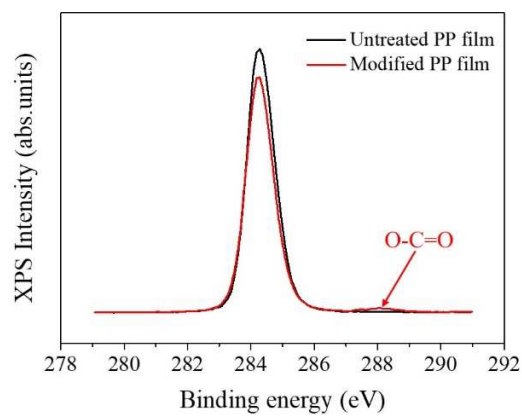


Fig. 6. XPS spectrum for PP film before and after microwave-assisted grafting with MAH in aqueous solution. Microwave power: 900 W, temperature: 80°C, time: 30min, MAH: 6g, APS: 0.6g.

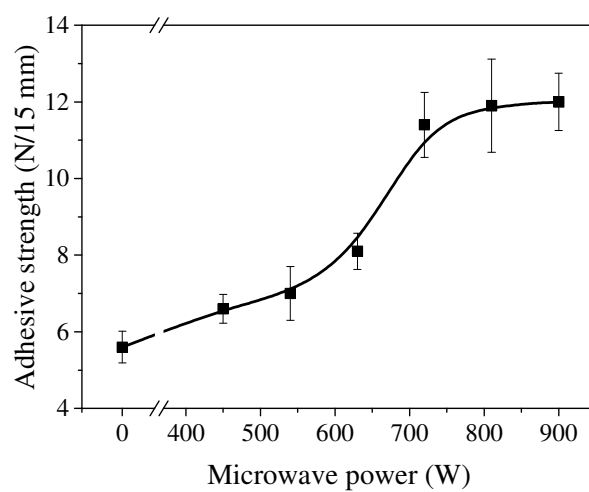


Fig. 7. Effect of microwave power on the adhesive strength between PP-g-MAH and PA6 double layer film. PP-g-MAH obtained under the following conditions: MAH: 6g, APS: 0.6g, temperature: 80°C, time: 30 min, xylene/water ratio: 1/40.

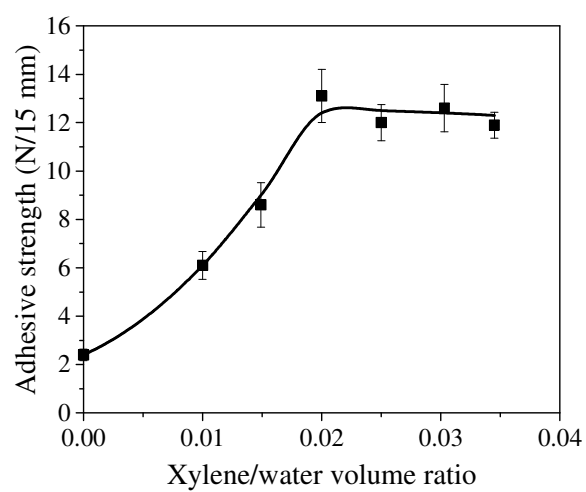


Fig. 8. Effect of the presence of xylene in water on the adhesive strength between the PP-g-MAH and PA6 layers. PP-g-MAH obtained under the following conditions: MAH: 6g, and APS: 0.6g, temperature: 80°C, time: 30 min, microwave power: 900W.



(a) xylene/water = 1/100



(b) xylene/water = 1/50



(c) xylene/water = 1/33

Fig. 9. Photographs showing the dyeing/swelling process of the PP film in water in the presence of various amounts of xylene at 80°C. Xylene/water ratio: (a) 1/100, (b) 1/50 and (c) 1/33. On the left: top views of the mixtures of dyed xylene and water before the immersion of a PP film, in the middle: top views of the mixtures of dyed xylene and water after the PP film is immersed and then taken out, on the right: dyed PP films.

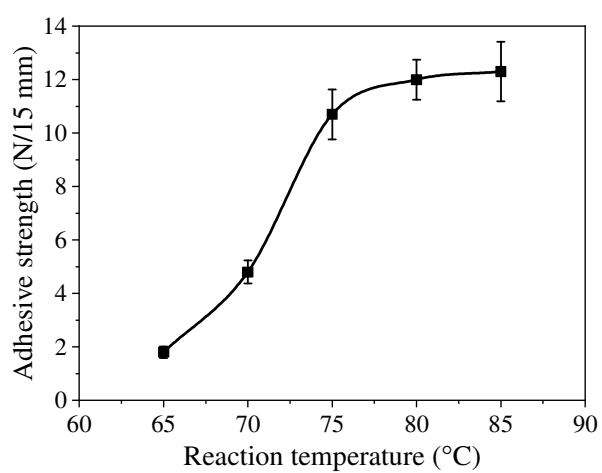


Fig. 10. Effect of temperature on the adhesive strength between the PP-g-MAH and PA6 layers.

PP-g-MAH obtained under the following conditions: MAH: 6g, APS: 0.6g, xylene/water: 1/40, time: 30 min, microwave power: 900W.

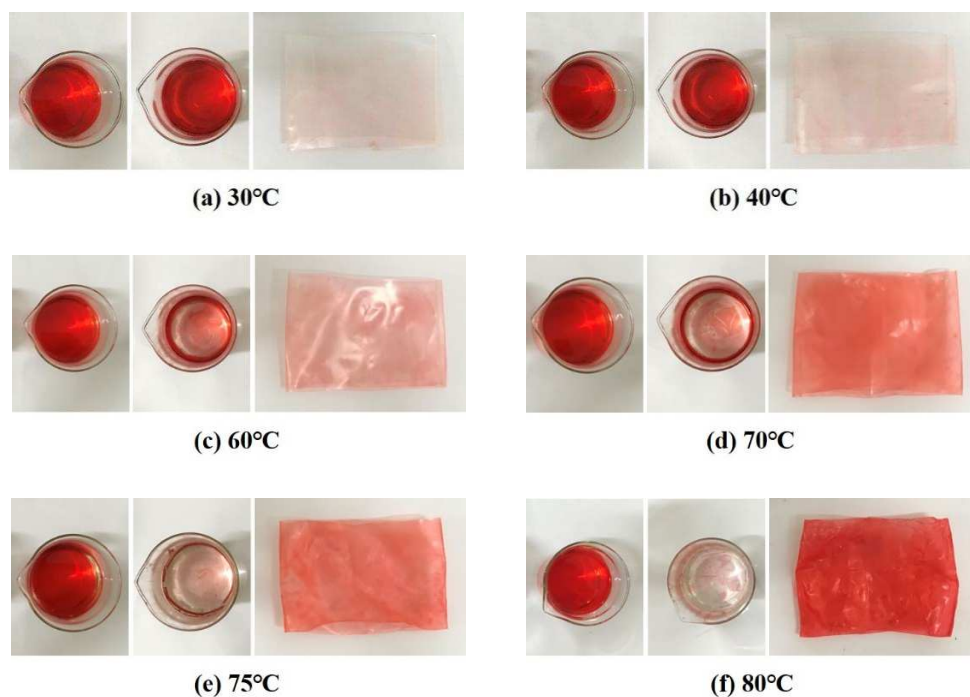


Fig. 11. Photographs showing the absorption/swelling process of the PP film in the mixture of xylene and water (xylene/water ratio: 1/40) at various temperatures: (a) 30°C, (b) 40°C, (c) 60°C, (d) 70°C, (e) 75°C, and (f) 80°C. On the left: top views of the mixtures of dyed xylene and water before the immersion of PP film, in the middle: top views of the mixtures of dyed xylene and water after the PP film is immersed and then taken out, on the right: dyed PP films.

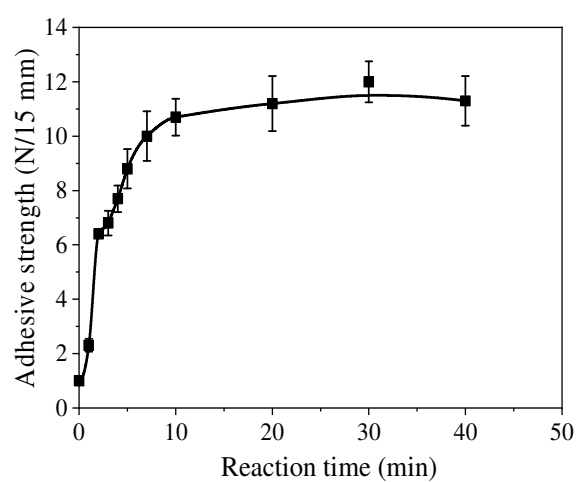


Fig. 12. Effect of time on the adhesive strength between the PP-g-MAH and PA6 layers. PP-g-MAH obtained under the following conditions: MAH: 6g, APS: 0.6g, xylene/water: 1/40, temperature: 80°C, microwave power: 900W.

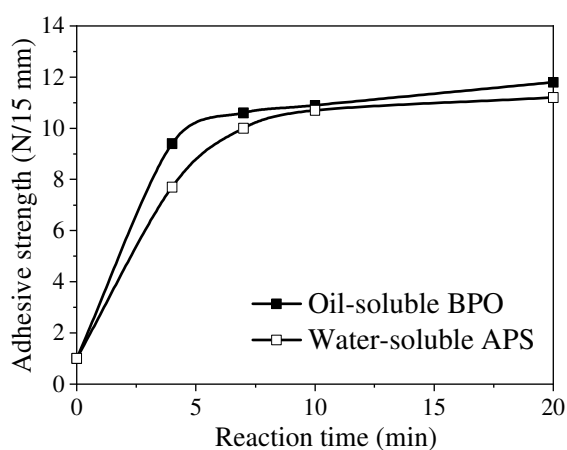


Fig. 13. Comparison between an oil-soluble BPO and a water-soluble APS as free radical initiators for the microwave-assisted free radical grafting of MAH onto the PP film surfaces. PP-g-MAH obtained under the following conditions: MAH (6.0g) dissolved in water (200mL), temperature: 80°C, microwave power: 900W. Note that BPO (only 0.01g) is dissolved in xylene (5mL) and then charged to water, whereas APS (0.6g) is dissolved in water in the presence of xylene (5mL).

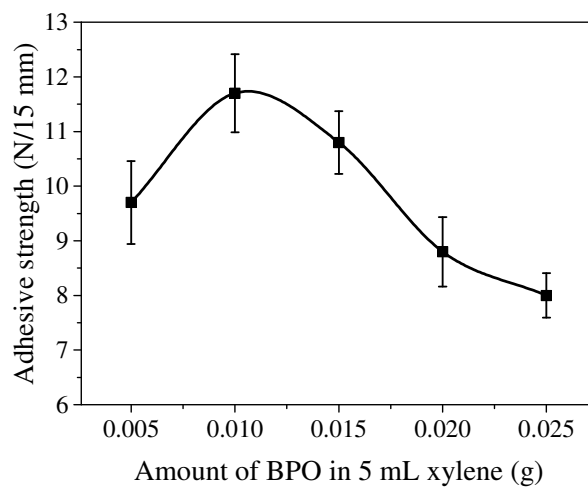


Fig. 14. Effect of the amount of BPO in xylene (5mL) on the adhesive strength between the PP-g-MAH and PA6 layers. PP-g-MAH obtained under the following conditions: MAH: 6g, xylene/water: 1/40, temperature: 80°C, time: 30min, microwave power: 900W.

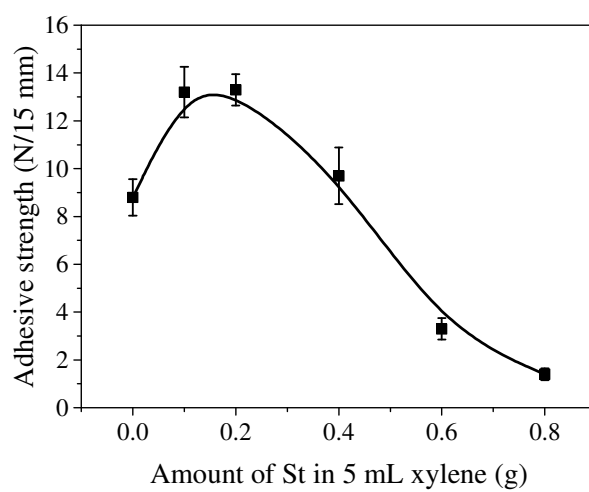
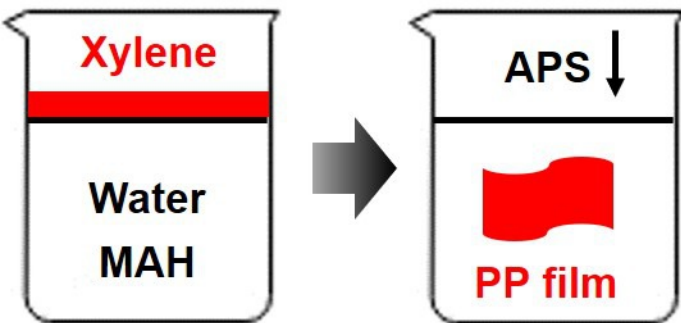
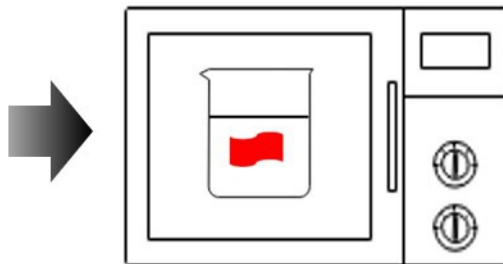


Fig. 15. Effect of adding St to xylene on the adhesive strength between the PP-g-MAH and PA6 films. PP-g-MAH obtained under the following conditions: Xylene/water: 1/40, MAH (6.0g) dissolved in water (200mL), BPO (0.02g) and St (various amounts) dissolved in xylene (5mL), temperature: 80°C, time: 30min, microwave power: 900W.

PP film surface swelling



Microwave-assisted grafting



side view of PP film

11.9 μm

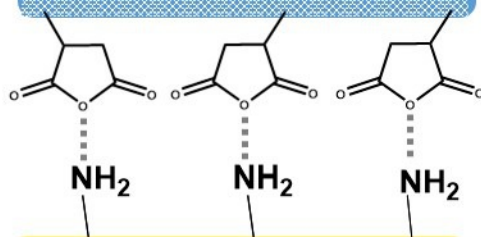
xylene

4 μm

SU3500 5.00kV 6.2mm x2.00k SE

20.0 μm

Modified PP film



PA6 film